Q1. JEE Main 2026 (21 January Shift 2)

Let $[\cdot]$ denote the greatest integer function and $f(x)=\lim_{n\to\infty}\frac{1}{n^3}\sum_{k=1}^{n}\left[\frac{k^2}{3^x}\right]$. Then $12\sum_{j=1}^{\infty}f(j)$ is equal to _____.

Q2. JEE Main 2026 (22 January Shift 2)

If $\lim_{x\to 0}\frac{e^{(a-1)x}+2\cos bx+(c-2)e^{-x}}{x\cos x-\log_e(1+x)}=2$, then $a^2+b^2+c^2$ is equal to :

(1) 5 (2) 9 (3) 3 (4) 7

Q3. JEE Main 2026 (21 January Shift 1)

Let $f:\mathbf{R}\to(0,\infty)$ be a twice differentiable function such that $f(3)=18, f'(3)=0$ and $f''(3)=4$. Then $\lim_{x\to 1}\left(\log_e\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right)$ is equal to :

(1) 1 (2) 18 (3) 2 (4) 9

Q4. JEE Main 2026 (28 January Shift 1)

The value of $\lim_{x\to 0}\frac{\log_e\left(\sec(ex)\cdot\sec\left(e^2x\right)\cdot\ldots\cdot\sec\left(e^{10}x\right)\right)}{e^2-e^{2\cos x}}$ is equal to

(1) $\frac{(e^{20}-1)}{2(e^2-1)}$ (2) $\frac{(e^{10}-1)}{2(e^2-1)}$ (3) $\frac{(e^{10}-1)}{2e^2(e^2-1)}$ (4) $\frac{(e^{20}-1)}{2e^2(e^2-1)}$

ANSWERS AND SOLUTIONS

1. 2 **2.** (4) **3.** (3) **4.** (1)

1. **(2)** Let $[\cdot]$ denote the greatest integer function. We have $f(x) = \lim_{n\to\infty} \frac{1}{n^3} \sum_{k=1}^{n} [\frac{k^2}{3^x}]$.

For large n: $\sum_{k=1}^{n} [\frac{k^2}{3^x}] \approx \sum_{k=1}^{n} \frac{k^2}{3^x} = \frac{1}{3^x} \cdot \frac{n(n+1)(2n+1)}{6} \approx \frac{n^3}{3 \cdot 3^x}$

Therefore: $f(x) = \lim_{n\to\infty} \frac{1}{n^3} \cdot \frac{n^3}{3 \cdot 3^x} = \frac{1}{3^{x+1}}$

Computing the infinite series:

$\sum_{j=1}^{\infty} f(j) = \sum_{j=1}^{\infty} \frac{1}{3^{j+1}} = \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \cdots$

$= \frac{1}{9} \cdot (1 + \frac{1}{3} + \frac{1}{9} + \cdots) = \frac{1}{9} \cdot \frac{1}{1-1/3} = \frac{1}{9} \cdot \frac{3}{2} = \frac{1}{6}$

Therefore: $12 \sum_{j=1}^{\infty} f(j) = 12 \cdot \frac{1}{6} = 2$

2.

(4) $\lim_{x\to 0} \frac{\left(1+(a-1)x+\frac{(a-1)^2x^2}{2!}\right)+2\left(1-\frac{b^2x^2}{2!}\right)+(c-2)\left(1-x+\frac{x^2}{2!}\right)}{x\left(1-\frac{x^2}{2!}\right)-\left(x-\frac{x^2}{2}\ldots\right)} = 2$

$\lim_{x\to 0} \frac{(1+2+c-2)+x(a-1-c+2)+x^2\left(\frac{(a-1)^2}{2}-b^2+\left(\frac{c-2}{2}\right)\right)}{\frac{x^2}{2}-\frac{x^3}{2!}+\ldots} = 2$

For which

$\because c + 1 = 0 \Rightarrow c = -1$

$\because a - c = -1 \Rightarrow a = -2$

$\because \frac{(a-1)^2}{2} - b^2 + \left(\frac{c-2}{2}\right) = 1$

$\frac{9}{2} - b^2 - \frac{3}{2} = 1 \Rightarrow b^2 = 2$

$a^2 + b^2 + c^2 = 4 + 2 + 1 = 7$

3.

(3) Let $T = \lim_{x\to 1} \left(\frac{f(x+2)}{f(3)}\right)^{\frac{18}{(x-1)^2}}$; 1^∞ form

$\Rightarrow T = e^{\lim_{x\to 1} \frac{18}{(x-1)^2}\left(\frac{f(x+2)-f(3)}{f(3)}\right)}$

$\Rightarrow T = e^{\lim_{x\to 1} \frac{18}{(x-1)^2}\left(\frac{f(x+2)-f(3)}{18}\right)}$

$\Rightarrow T = e^{\lim_{x\to 1}\left(\frac{f(x+2)-f(3)}{(x-1)^2}\right)}$ $\frac{0}{0}$ form, apply L'pital

$\Rightarrow T = e^{\lim_{x\to 1} \frac{f'(x+2)}{2(x-1)}}$; $\frac{0}{0}$ form, apply L'pital

$\Rightarrow T = e^{\lim_{x\to 1} \frac{f''(x+2)}{2}} = e^{\frac{4}{2}} = e^2$

$\Rightarrow \log_e(T) = 2$

4. **(1)** Numerator: $\log_e(\sec(ex)\sec(e^2x)\cdots\sec(e^{10}x)) = \sum_{k=1}^{10}\log_e(\sec(e^kx))$.

Using $\sec(u) \approx 1 + \frac{u^2}{2}$ for small u, we get $\log_e(\sec(e^kx)) \approx \frac{e^{2k}x^2}{2}$.

Thus the numerator is $\approx \frac{x^2}{2}\sum_{k=1}^{10} e^{2k} = \frac{x^2e^2(e^{20}-1)}{2(e^2-1)}$.

Denominator: Using $\cos x \approx 1 - \frac{x^2}{2}$, we have $e^{2\cos x} \approx e^2(1-x^2)$, so $e^2 - e^{2\cos x} \approx e^2x^2$.

Therefore: $\lim_{x\to 0} = \frac{\frac{x^2e^2(e^{20}-1)}{2(e^2-1)}}{e^2x^2} = \frac{e^{20}-1}{2(e^2-1)}$.